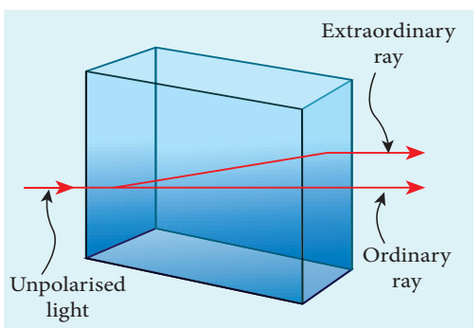
This phenomenon is also exhibited by crystals like quartz, mica etc.

When a dot of ink on a sheet of paper is viewed through a calcite crystal, two images will be seen. On rotating the crystal, one image remains stationary and the other rotates around it. The stationary image *O* is produced by ordinary rays which obey the laws of refraction. The rotating image *E* is produced by extraordinary rays which do not obey the laws of refraction. The extraordinary ray is found to be plane polarised. Inside a double refracting crystal the ordinary ray travels with same velocity in all directions and the extra ordinary ray travels with different velocities in all directions. A point source inside the crystal produces spherical wavefront for ordinary ray and elliptical wavefront for extraordinary ray. Inside the crystal, there is a particular direction in which both the rays travel with same velocity. This direction is called known as *optic axis*. Along the optic axis, the refractive index is same for both the rays and there is no double refraction along this axis.



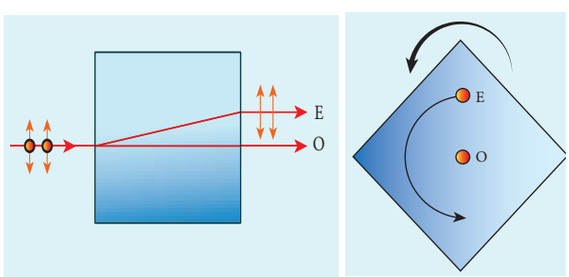


Figure 7.33 Double refraction

7.5.6 Types of optically active crystals

Crystals like calcite, quartz, tourmaline and ice which have only one optic axis are called uniaxial crystals.

Crystals like mica, topaz, selenite and aragonite which have two optic axes are called biaxial crystals.

7.5.7 Nicol prism

Nicol prism is an optical device which forms a part of many optical instruments both for producing plane polarised light and also analysing. The construction of a Nicol prism is based on the phenomenon of double refraction. It was designed by William Nicol in 1828.

Nicol prism is a calcite crystal which has a length three times its breadth and angles 72° and 108°. It is cut into two halves along the diagonal as shown in Figure 7.34. The two halves are pasted together with a layer of *canada balsam*, a transparent cement.

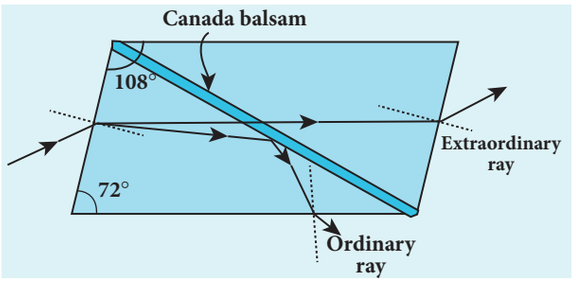


Figure 7.34 Nicol Prism

Let us consider a ray of unpolarised light from a monochromatic source is incident on the Nicol prism. The double refraction takes place and the ray is split into ordinary and extraordinary rays. They travel in different directions with different velocities. For monochromatic sodium light the refractive index of the

UNIT-7(XII-Physics_Vol-2)_new.indd 86 21-07-2020 16:36:24